

Effectful Computations with Future Conditions: A Monadic Approach to Temporal Specification

Functional Pearl

AUTHOR NAME, Institution, Country

Pre- and post-conditions are the standard vocabulary for modular program specification, yet they are silent about obligations that span multiple calls. In real Haskell code such obligations are ubiquitous: a successful TLS handshake must eventually be followed by `bye` before the context is discarded; a taken `MVar` must eventually be put back on *every* code path, including those interrupted by asynchronous exceptions; a `beginTx` must eventually reach `commit` or `rollback`; a submitted job must complete with probability at least p . No single pre/post pair can express these properties, because the balancing operation lies in an unknown future context.

We present *Pledge*, a Haskell library that fills this gap by augmenting the classical contract model with *future conditions*: annotations that propagate obligations forward through a computation until they are fully discharged. Future conditions are carried as a data-dependent field `future :: a → eff` in the *Pledge* monad alongside pre- and post-conditions. Value-dependence is essential: it lets an obligation name the particular resource handle the operation returns — requiring, for instance, that the *specific* context produced by `tlsHandshake` must eventually have `bye` called on it. When operations are sequenced via monadic `bind`, the future condition of the first computation is partially discharged by the postcondition of its continuation through a *subtraction* operator; the residual is then conjoined with the continuation’s own future condition. Annotations are per-operation; the monad handles all propagation and discharge automatically.

The design is parameterised over a *Composable algebra* requiring concatenation ($\diamond$), conjunction ($\wedge$), and subtraction ($\setminus$), together with identity elements `empty` and `universe`. We present four instances: trace protocols via extended regular expressions (RE, with Brzowski-derivative subtraction and a direct embedding of LTL_f); trace protocols enriched with Presburger arithmetic bounds (GRE, discharged by an SMT solver); quantitative obligations via semiring-weighted regular expressions (WRE, capturing probabilistic and min-cost scenarios); and heap-ownership obligations via separation logic (SL, with magic wand as subtraction). All four instances share a single monadic `bind` formula, showing that diverse obligation domains — trace-based, arithmetic, quantitative, and heap-ownership — are uniformly expressible within one algebraic interface.

CCS Concepts: • **Software and its engineering** → **Functional languages**; *Data types and structures*; *Software verification and validation*.

Additional Key Words and Phrases: temporal specification, monadic effects, regular expressions, complement, Brzowski derivatives, Presburger arithmetic, semiring weights, separation logic, resource lifecycle, Haskell

ACM Reference Format:

Author Name. 2026. Effectful Computations with Future Conditions: A Monadic Approach to Temporal Specification: Functional Pearl. *Proc. ACM Program. Lang.* 1, HASSELL (May 2026), 25 pages.

1 Introduction

Pre- and post-conditions are the workhorse of modular program specification. A pre-condition constrains the state in which an operation may be invoked; a post-condition describes what holds

Author’s Contact Information: Author Name, Institution, City, Country, author@institution.edu.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

© 2026 Copyright held by the owner/author(s). Publication rights licensed to ACM.

ACM 2475-1421/2026/5-ART

<https://doi.org/>

immediately after it returns. Yet a great many correctness properties are not local to a single call boundary: they *span* a sequence of operations, with arbitrary computation in between.

The problem in real Haskell code. Consider three scenarios from the Haskell ecosystem.

TLS lifecycle. A successful handshake must eventually be followed by `bye` before the context is discarded. A post-condition on handshake records success, but cannot enforce that `bye` will be called, because the discharge point lies in an unknown future context. `bracket` sidesteps this by lexically scoping the context, but cannot express richer message-exchange obligations or handle contexts passed across function boundaries.

MVar discipline. `takeMVar` removes a value, leaving the variable empty; the caller must restore it with `putMVar` on every exit path including `async-exception` paths. A post-condition records the empty state but cannot require that a future `putMVar` on the *same* variable will occur.

Database savepoints. `transactionSave` opens a fresh transaction that must itself be committed or rolled back; pre/post-conditions on the call cannot carry this obligation forward.

In each case the obstacle is identical: the `postcondition` cannot name what comes next because the continuation is unknown at specification time.

Existing remedies and their limits. `bracket` lexically scopes cleanup but cannot propagate obligations beyond the lexical scope. `ResourceT` [Snoyman 2011] and `Acquire/Managed` thread cleanup actions through every `bind` but model only the cleanup half of an obligation, discharging it uniformly at scope exit rather than by application-specific operations. Linear Haskell [Bernardy et al. 2018] enforces “consume exactly once” but cannot express branching discharge obligations or quantitative commitments. Parameterised and indexed monads [Atkey 2009] encode protocol states at the type level but require `RebindableSyntax` or `QualifiedDo`, produce opaque type errors, and are confined to a single domain per design.

Our approach. We propose the `Pledge` monad as a lightweight, compositional mechanism for specifying such obligations, requiring no language extensions beyond standard Haskell type classes. A value of type `Pledge eff a` carries three annotations: a precondition `pre :: eff`, a postcondition `post :: eff`, and a data-dependent future condition `future :: a → eff`. Because `future` ranges over the return value, an obligation can name the actual resource handle produced at runtime rather than one fixed at specification time: `tlsHandshake` can declare that *whichever* `Context` it returns must eventually receive `bye`. Monadic `bind` partially discharges the first computation’s future condition against the continuation’s postcondition via subtraction; the residual is conjoined with the continuation’s own future condition. The user annotates individual operations only; the monad propagates obligations automatically.

The `Composable` typeclass abstracts the specification algebra into five operations: concatenation ($\diamond$), conjunction ($\wedge$), subtraction ($\setminus$), and their respective identities `empty` and `universe`. It admits four independent instances:

- (*Trace protocol*, RE) Extended regular expressions over events, with complement as a first-class constructor. Subtraction is implemented via Brzowski derivatives; LTL_f operators embed directly as RE terms without automaton construction. This instance handles `takeMVar/putMVar` discipline, TLS handshake/bye pairing, transaction lifecycle, and lock protocols.
- (*Arithmetic bounds*, GRE) Each RE is paired with a Presburger arithmetic predicate over program variables, checked by an SMT solver (Z3 via SBV). A single annotation simultaneously enforces a protocol ordering *and* a numeric bound, catching, for example, counter overflow before any execution takes place.
- (*Quantitative obligation*, WRE) Boolean membership is generalised to a semiring weight. The probability semiring captures obligations of the form “this job completes with probability at

least p "; the tropical semiring gives minimum-cost scheduling obligations. A weight of szero in the residual signals an unfulfilled quantitative commitment.

- (*Heap ownership*, SL) Separation-logic heap predicates, with separating conjunction as sequential composition and magic wand as subtraction. Future conditions describe what heap state must hold when the computation completes; a leaked `Cell` in the residual post names the unreleased ownership precisely.

That all four share a single monadic bind formula is not a coincidence: trace-based, arithmetic, quantitative, and heap-ownership obligations are structurally identical at the level of the `Composable` algebra.

Contributions.

- (1) We define the `Pledge` monad (§3), parameterised over a `Composable` algebra, with `pre :: eff`, `post :: eff`, and a data-dependent future `a → eff` field, and verify that the monad laws hold for the $(ret, post, future)$ triple (§9).
- (2) We instantiate the algebra to extended regular expressions over events with complement, using Brzozowski derivatives to implement subtraction (§2.1), and show that LTL_f operators embed directly as RE terms (§2.3).
- (3) We extend the RE instance to a guarded form (GRE, §5), pairing each RE with a Presburger arithmetic predicate over program variables, evaluated by an SMT solver (Z3 via SBV) at membership-check time.
- (4) We provide a weighted RE instance (WRE, §6), parameterised over a semiring, that generalises Boolean membership to a quantitative weight, recovering probabilistic and min-cost future conditions as special cases.
- (5) We provide a fourth, independent instantiation to separation-logic heap predicates (§7), demonstrating that the `Composable` interface extends beyond trace-based specifications.
- (6) We demonstrate the approach on ten use cases across all four instances (§8), showing that resource leaks, ordering violations, arithmetic overflows, probabilistic obligations, and heap-ownership errors all surface as non-trivial residual annotations.

2 Background

2.1 Extended Regular Expressions over Events

Events are the atomic units of observation; each carries a name and a list of typed arguments.

```
data Term = Str String | Num Int | List [Term]

data Event = Atom String [Term] -- concrete event, e.g. Atom "send" [Num 1]
           | Wildcard           -- pattern matching any single event

data RE = Bot | Epsilon | Single Event
        | Seq RE RE | Or RE RE | And RE RE | Star RE | Not RE
```

The specification language is a regular expression type that adds complement as a first-class constructor:

$$r ::= \emptyset \mid \varepsilon \mid e \mid r_1 \cdot r_2 \mid r_1 \vee r_2 \mid r_1 \wedge r_2 \mid r^* \mid \neg r$$

Here e is a `Single` Event that uses `Wildcard` as a pattern for any event, $\wedge$ is intersection (derived: $r_1 \wedge r_2 \triangleq \neg(\neg r_1 \vee \neg r_2)$), and $\neg r$ is the complement with respect to Σ^* .

Five derived forms are used throughout:

$\text{anything} \triangleq \neg\emptyset$	(the universal language Σ^*)
$\text{finally}(e) \triangleq \text{anything} \cdot e \cdot \text{anything}$	(e must occur eventually)
$\text{globally}(e) \triangleq e^*$	(e at every step)
$\text{never}(e) \triangleq \neg\text{finally}(e)$	(e must never occur)
$\text{noUntil}(e, g) \triangleq \neg((\Sigma \setminus \{g\})^* \cdot e \cdot \text{anything})$	(e must not occur before g)

$\text{noUntil}(e, g)$ is nullable (the empty continuation satisfies it) and its derivative with respect to g is Σ^* , meaning once g occurs the restriction on e is lifted. This is the key combinator for *conditional* safety: e is forbidden until g re-enables it.

2.2 Algebraic Complement and Brzowski Derivatives

The Brzowski derivative [Brzowski 1964] $\partial_a(r)$ computes the language of continuations after reading event a . The key law for complement is:

$$\partial_a(\neg r) = \neg(\partial_a(r)) \quad (1)$$

The nullability function $\nu(r)$ ($= \text{True}$ iff $\varepsilon \in L(r)$) satisfies $\nu(\neg r) = \neg \nu(r)$. Together, equations (1) and the De Morgan laws

$$\neg(r_1 \vee r_2) = \neg r_1 \wedge \neg r_2 \quad \neg(r_1 \wedge r_2) = \neg r_1 \vee \neg r_2$$

allow complement to be propagated and simplified *purely algebraically*, without building a DFA. The normaliser additionally applies $\neg\neg r = r$, $\neg\emptyset = \emptyset$, and $\neg\neg\emptyset = \emptyset$.

In Haskell:

```

nullable :: RE → Bool
nullable (Not r)      = not (nullable r)      -- nu(Not r) = not(nu(r))
nullable (Star _)     = True
nullable (Seq r1 r2)  = nullable r1 && nullable r2
nullable (Or r1 r2)   = nullable r1 || nullable r2
nullable (And r1 r2)  = nullable r1 && nullable r2
nullable Epsilon      = True
nullable _            = False

derivative :: Event → RE → RE
derivative e (Not r)   = Not (derivative e r)  -- d_a(Not r) = Not(d_a(r))
derivative e (Single p) = if matchEvent e p then Epsilon else Bot
derivative e (Seq r1 r2)
  | nullable r1 = Or (Seq (derivative e r1) r2) (derivative e r2)
  | otherwise  = Seq (derivative e r1) r2
derivative e (Or r1 r2) = Or (derivative e r1) (derivative e r2)
derivative e (And r1 r2) = And (derivative e r1) (derivative e r2)
derivative e (Star r)   = Seq (derivative e r) (Star r)
derivative _ _         = Bot

```

The `firstWith` function returns the set of events (drawn from a supplied finite alphabet Σ_0) that can begin a word in $L(r)$, driving the subtraction algorithm. The complement case requires

particular care. Naively writing $\text{first } (\text{Not } r) = \text{first } r$ is *unsound*: $L(\neg r)$ can begin with events *not* in $\text{first}(r)$, so the naive rule explores the wrong direction. The correct characterisation is:

$$e \in \text{first}(\neg r, \Sigma_0) \iff e \in \Sigma_0 \wedge [\partial_e(r)] \neq \neg\emptyset \quad (2)$$

That is, e can open a word in $L(\neg r)$ iff after consuming e , the residual $\partial_e(r)$ does *not* cover the whole future; there exists some continuation that stays outside $L(r)$. The alphabet Σ_0 is collected from an RE via atoms, which recursively gathers all concrete (**Atom**) events, excluding **Wildcard**.

```
atoms :: RE → [Event]
atoms (Single Wildcard) = []
atoms (Single e)         = [e]
atoms (Seq r1 r2)        = atoms r1 `union` atoms r2
atoms (Or r1 r2)         = atoms r1 `union` atoms r2
atoms (And r1 r2)        = atoms r1 `union` atoms r2
atoms (Star r)           = atoms r
atoms (Not r)            = atoms r
atoms _                  = []

firstWith :: [Event] → RE → [Event]
firstWith _ Bot          = []
firstWith _ Epsilon      = []
firstWith _ (Single e)   = [e]
firstWith alph (Seq r1 r2)
  | nullable r1           = firstWith alph r1 `union` firstWith alph r2
  | otherwise             = firstWith alph r1
firstWith alph (Or r1 r2) = firstWith alph r1 `union` firstWith alph r2
firstWith alph (And r1 r2) = [e | e <- firstWith alph r1,
                                   e `elem` firstWith alph r2]
firstWith alph (Star r)   = firstWith alph r
firstWith alph (Not r)    = [e | e <- alph,
                                   not (isTotal (normalize (derivative e r)))]

where isTotal (Not Bot) = True; isTotal _ = False
```

The function $\text{first } r = \text{firstWith } (\text{atoms } r) \text{ } r$ is a convenient wrapper for standalone use; inside subtraction both operands contribute to the alphabet (see §4).

Illustrative cases. Table 1 gives representative evaluations of firstWith with **Not**. The first four rows confirm that complement does not exclude any alphabet event when the derivative is non-total. The double-negation row shows correct restoration of the original first set: $\partial_b(\neg a) = \Sigma^*$ is total, so b is filtered out. In $\neg(a \cdot \Sigma^*)$, $\partial_a(a \cdot \Sigma^*)$ normalises to Σ^* , excluding a ; only b survives. Similarly, $\neg(a \cdot \Sigma^* \vee b \cdot \Sigma^*)$ excludes both a and b , leaving only c , while the De Morgan row verifies that $\neg(\neg a \wedge \neg b) = a \vee b$ yields $\{a, b\}$.

2.3 Embedding LTL_f

Because complement is a first-class constructor, the standard LTL_f operators translate directly into RE, with no automaton construction needed. Table 2 gives the full mapping.

The complement operator does the heavy lifting: $G \phi \triangleq \neg F \neg \phi$ unfolds to $\neg(\neg\emptyset \cdot \neg\llbracket\phi\rrbracket)$ using two applications of $\neg$; $F \phi$ and $X \phi$ need only concatenation. The U case uses $\text{step}(\phi)$, the length-1 RE for a single event satisfying ϕ , which is defined only when ϕ is propositional.

Table 1. Selected `firstWith` evaluations with **Not**. Σ_0 is the explicitly supplied alphabet. Rows marked $\dagger$ produce a *strict* subset of Σ_0 .

RE r	Σ_0	<code>firstWith</code> Σ_0 ($\neg r$)	Rationale
a	$\{a, b\}$	$\{a, b\}$	$\partial_a(a) = \varepsilon, \partial_b(a) = \emptyset$; neither Σ^*
$a \cdot b$	$\{a, b, c\}$	$\{a, b, c\}$	all derivatives non- Σ^*
$a \vee b$	$\{a, b, c\}$	$\{a, b, c\}$	$\partial_c(a \vee b) = \emptyset$, still non- Σ^*
a^*	$\{a, b\}$	$\{a, b\}$	$\partial_a(a^*) = a^*$, non- Σ^* ; b starts $[b]$
$\neg a$ (double neg.)	$\{a, b\}$	$[a]$	$\partial_b(\neg a) = \Sigma^* \Rightarrow b$ excluded †
$a \cdot \Sigma^*$	$\{a, b\}$	$[b]$	$\partial_a(a \cdot \Sigma^*) = \Sigma^* \Rightarrow a$ excluded †
$a \cdot \Sigma^* \vee b \cdot \Sigma^*$	$\{a, b, c\}$	$[c]$	both a, b excluded; only c avoids †
$\neg a \wedge \neg b$	$\{a, b, c\}$	$\{a, b\}$	De Morgan: $= \neg(a \vee b)$; ∂_c is $\Sigma^{*\dagger}$

Table 2. LTL_f to RE translation. $\Sigma^1 \triangleq \text{Single Wildcard}$, $\neg \emptyset \triangleq \neg \emptyset$.

LTL formula	RE $\llbracket \cdot \rrbracket$
$\top$	$\neg \emptyset$
$\perp$	$\emptyset$
e	$\{e\}$
$\neg \phi$	$\neg \llbracket \phi \rrbracket$
$\phi \wedge \psi$	$\llbracket \phi \rrbracket \cap \llbracket \psi \rrbracket$
$\phi \vee \psi$	$\llbracket \phi \rrbracket \cup \llbracket \psi \rrbracket$
$X \phi$	$\Sigma^1 \cdot \llbracket \phi \rrbracket$
$F \phi$	$\neg \emptyset \cdot \llbracket \phi \rrbracket$
$G \phi$	$\neg(\neg \emptyset \cdot \neg \llbracket \phi \rrbracket)$
$\phi U \psi$	$\text{step}(\phi)^* \cdot \llbracket \psi \rrbracket$ (ϕ propositional)

3 The Pledge Monad

3.1 The Composable Class

We abstract the specification algebra via a type class:

```

class Composable a where
  concatenation :: a → a → a  -- (◊), unit: empty
  conjunction  :: a → a → a  -- (∧), unit: universe
  empty        :: a           -- identity for ◊ (varepsilon)
  universe     :: a           -- identity for ∧ (neg-emptyset)
  subtraction  :: a → a → a  -- (\\)

```

The five operations form exactly the interface required by the `bind` formula. `concatenation` sequences effects; `conjunction` combines simultaneous constraints; `subtraction` computes the residual obligation (the Brzozowski quotient for the RE instance). `empty` and `universe` are the respective identity elements and appear in `pure`.

3.2 The Pledge Type

The future field is *data-dependent*: it is a function of the return value a , allowing obligations to refer to the specific resource handle that an operation produces. The helper `evalFuture` evaluates this function at the computation's own return value.

```
data Pledge eff a = Pledge
  { ret    :: a
  , pre    :: eff          -- what must have occurred before
  , post   :: eff          -- what this computation produces
  , future :: a → eff      -- obligation indexed by the return value
  }

evalFuture :: Pledge eff a → eff
evalFuture e = future e (ret e)
```

3.3 Monad Instances

`pure` introduces a value with no effect and no obligation:

```
pure x = Pledge { ret = x, pre = universe, post = empty, future = \_ → universe }
```

The bind operator sequences two computations; write $e_2 = f(\text{ret}(e))$.

```
e >>= f = let fe = f (ret e) in Pledge
  { ret    = ret fe
  , pre    = pre e ∧ (post e \ \ pre fe)
  , post   = post e ∨ post fe
  , future = \_ → (post fe \ \ future e (ret e)) ∧ future fe (ret fe)
  }
```

Future condition. Each future function is applied to the corresponding `ret` before the obligation is propagated:

$$\text{future}(e \gg f) = (\text{post}(e_2) \setminus \text{future}(e)(\text{ret}(e))) \wedge \text{future}(e_2)(\text{ret}(e_2)) \quad (3)$$

This removes from e 's future obligation (evaluated at e 's return value) those traces that e_2 's postcondition satisfies, then intersects the residual with e_2 's own future obligation (evaluated at e_2 's return value). When the result normalises to $\neg\emptyset$, all obligations are discharged.

Precondition.

$$\text{pre}(e \gg f) = \text{pre}(e) \wedge (\text{post}(e) \setminus \text{pre}(e_2)) \quad (4)$$

The term $\text{post}(e) \setminus \text{pre}(e_2)$ is the *residual precondition*: the part of e_2 's precondition not already discharged by e 's postcondition. The combination uses $\wedge$ (intersection): both $\text{pre}(e)$ and the residual must hold *simultaneously* in the same input history.

When $\text{post}(e)$ does not cover $\text{pre}(e_2)$ at all, the residual is $\emptyset$, flagging a precondition violation. When $\text{post}(e)$ exactly satisfies $\text{pre}(e_2)$, the residual is ε and the combined precondition becomes $\text{pre}(e) \wedge \varepsilon$. If $\text{pre}(e) = \neg\emptyset$ (nullable), this reduces to ε ; if $\text{pre}(e) = \{a\}$ (non-nullable), the intersection $\{a\} \wedge \varepsilon = \emptyset$ flags the contradiction immediately. This behaviour is intentional: incompatible constraints collapse to the empty intersection rather than silently discarding one side.

3.4 Intuition via an Example

`takeMVar` returns the handle of the taken `MVar` as its return value; the future condition is data-dependent, referring to whatever handle was actually returned. `putMVar` asserts that `putMVar(v)` must not recur until `takeMVar(v)` is called again, preventing double-put.

```
type MVar = Int
```

```
takeMVar :: MVar → Pledge RE MVar
```

```
takeMVar v = Pledge
```

```
  { ret = v, pre = universe
  , post  = Single (Atom "takeMVar" [Num v])
  , future = \r → finally (Atom "putMVar" [Num r]) } }
```

```
putMVar :: MVar → Pledge RE ()
```

```
putMVar v = Pledge
```

```
  { ret = ()
  , pre  = Single (Atom "takeMVar" [Num v])
  , post = Single (Atom "putMVar" [Num v])
  , future = \_ → noUntil (Atom "putMVar" [Num v]) (Atom "takeMVar" [Num v]) } }
```

Good program: `v <- takeMVar 1; putMVar v`. After `takeMVar 1` ($ret = 1$):

$$post = takeMVar(1), \quad future = finally(putMVar(1)), \quad pre = \neg\emptyset$$

Binding `putMVar 1` (with $pre = takeMVar(1)$, $post = putMVar(1)$):

Precondition: $post(takeMVar(1)) \setminus pre(putMVar(1)) = takeMVar(1) \setminus takeMVar(1) = \varepsilon$. So $pre = \neg\emptyset \wedge \varepsilon = \varepsilon$. ✓

Future: $putMVar(1) \setminus finally(putMVar(1)) = \neg\emptyset$. Intersect with $future(putMVar(1)) = noUntil(putMVar(1), result noUntil(putMVar(1), takeMVar(1)))$. ✓ (nullable: the program may end here; no double-put obligation is pending.)

Bad: `putMVar 1` alone. `pure ()` has $post = \varepsilon$. $takeMVar(1) \setminus \varepsilon = \emptyset$. So $pre = \emptyset$. ✗

Bad: `missingPut (takeMVar 1 >> takeMVar 2 >> putMVar 1)`. The future of `takeMVar 2` is `finally(putMVar(2))`, which is not discharged by `putMVar 1`, so the residual future still contains `finally(putMVar(2))`, naming the blocked `MVar` precisely.

Bad: `doublePutImmediate (v <- takeMVar 1; putMVar v; putMVar v)`. After the first `putMVar 1`, the future is `noUntil(putMVar(1), takeMVar(1))`. The second `putMVar 1` posts `putMVar(1)`; consuming `putMVar(1)` in `noUntil(putMVar(1), takeMVar(1))` yields $\emptyset$, so the combined future is $\emptyset$. ✗

4 The RE Instance

The `Composable RE` instance maps the algebra operations onto `RE` constructors and implements subtraction via derivatives:

```
instance Composable RE where
```

```
  concatenation = Seq
```

```
  conjunction  = And
```

```
  empty        = Epsilon
```

```
  universe     = anything    -- = Not Bot
```

```
  subtraction Epsilon r2 = r2
```

```
  subtraction (Not Bot) _ = Epsilon    -- Sigma* \ r = eps (trivially satisfied)
```

```
  subtraction r1 r2 =
```



```

let alph  = atoms r1 `union` atoms r2  -- combined alphabet
    evts   = firstWith alph r1
    step e = subtraction (normalize (derivative e r1))
                      (normalize (derivative e r2))
in foldr Or Bot (map step evts)

```

```

normalize :: RE → RE  -- RE-specific; not part of Composable
normalize r = {- ... algebraic simplification ... -}

```

The second clause handles the universal case: $\neg \emptyset \setminus r = \varepsilon$ for any r . The general clause collects the alphabet from *both* operands before calling `firstWith`, ensuring events appearing only in r_2 are still considered under complement via Equation (2).

`normalize` is defined outside the instance and applies the following RE-specific laws, which exploit the complement constructor:

Law	Applied to
$\varepsilon \cdot r = r, r \cdot \varepsilon = r$	Seq
$\emptyset \vee r = r, r \vee \neg \emptyset = \neg \emptyset$	Or
$\emptyset \wedge r = \emptyset, r \wedge \neg \emptyset = r$	And
$\neg \neg r = r$	Not
$\neg(r_1 \vee r_2) = \neg r_1 \wedge \neg r_2$	Not
$\neg(r_1 \wedge r_2) = \neg r_1 \vee \neg r_2$	Not
$\neg \emptyset = \neg \emptyset, \neg \neg \emptyset = \emptyset$	Not
$\emptyset^* = \varepsilon, \varepsilon^* = \varepsilon$	Star

5 The GuardedRE Instance

The RE instance reasons about event traces but is entirely silent about arithmetic state. The GRE instance closes this gap: a guarded regular expression pairs a trace RE with a Presburger arithmetic predicate over a concrete heap, so that a single annotation can simultaneously enforce a protocol ordering *and* an arithmetic bound.

5.1 The GuardedRE Type

```

data GuardedRE = GuardedRE PPred RE
    deriving (Eq)

```

A state (heap, trace) satisfies **GuardedRE** p r iff

$$\text{heap} \models p \quad \wedge \quad \text{trace} \in L(r).$$

The two sides are independent: p is a static constraint on the heap, while r advances through Brzozowski derivatives as events arrive.

Two smart constructors lift a pure RE or a pure Presburger predicate:

```

fromRE    :: RE      → GuardedRE
fromRE r   = GuardedRE PTrue r      -- no heap constraint

fromPPred :: PPred → GuardedRE
fromPPred p = GuardedRE p universe  -- accept any trace

```

5.2 The Composable GuardedRE Instance

```
instance Composable GuardedRE where
  concatenation (GuardedRE p1 r1) (GuardedRE p2 r2) =
    GuardedRE (normalizePPred (PAnd p1 p2)) (normalize (Seq r1 r2))
  conjunction (GuardedRE p1 r1) (GuardedRE p2 r2) =
    GuardedRE (normalizePPred (PAnd p1 p2)) (normalize (And r1 r2))
  empty = GuardedRE PTrue Epsilon
  universe = GuardedRE PTrue (Not Bot) -- = anything
  subtraction (GuardedRE p1 r1) (GuardedRE p2 r2) =
    GuardedRE (normalizePPred (PAnd p1 p2))
      (normalize (reSubtraction r1 r2))
```

Both concatenation and subtraction conjoin both heap predicates: the residual must respect the constraints from both operands. The RE component uses the same reSubtraction and normalize as the plain RE instance.

5.3 Presburger Normalisation

Composing multiple **GuardedRE** values accumulates a tree of **PAnd** nodes. **normalizePPred** flattens this tree, removes duplicates, substitutes equalities (e.g. $h[a] = k \Rightarrow$ replace **ValAt** a by k throughout), evaluates literal comparisons, and replaces $\neg \text{true}$ with the canonical **false** constructor **PFalse**:

```
data PPred
  = PTrue | PFalse -- canonical true/false
  | PLt PExpr PExpr | ...
  | PNot PPred | PAnd PPred PPred

normalizePPred :: PPred → PPred
normalizePPred p = rebuildAnd (substEqs eqs atoms)
  where
    atoms = flattenAnd (normStep p)
    eqs = [ (a, k) | PEq (ValAt a) (Lit k) <- atoms ]
```

The introduction of **PFalse** prevents the solver from being invoked with expressions such as $(\neg \text{true}) \wedge \dots$ that are syntactically unsatisfiable; these normalise immediately to **PFalse** without any solver call.

5.4 Membership Checking

Checking whether a concrete (*heap*, *trace*) satisfies a **GuardedRE** involves two steps: fold **deriveGuarded** over the trace (advancing only the RE side), then call **nullableGuarded** which checks RE nullability and invokes the SBV/Z3 backend on the instantiated Presburger predicate.

```
deriveGuarded :: Event → GuardedRE → GuardedRE
deriveGuarded e (GuardedRE p r) = GuardedRE p (normalize (derivative e r))

nullableGuarded :: Map Addr Int → GuardedRE → IO Bool
nullableGuarded heap (GuardedRE p r)
  | not (nullable r) = return False
  | otherwise = do
    result <- checkPPred (instantiate heap p)
    return $ case result of { Satisfied _ → True; _ → False }
```

```
checkGuarded :: Map Addr Int → [Event] → GuardedRE → IO Bool
checkGuarded heap trace g =
  nullableGuarded heap (foldl (flip deriveGuarded) g trace)
```

5.5 Example: Bounded Counter

A bounded counter stored at heap address a with maximum value M must follow the protocol $\text{init} \cdot (\text{inc} \mid \text{dec})^* \cdot \text{snapshot}$. Increment is additionally guarded by $h[a] < M$ (no overflow); decrement by $h[a] > 0$ (no underflow).

```
initCounter :: Addr → Pledge GuardedRE ()
initCounter a = Pledge
  { ret = ()
  , pre = fromRE universe
  , post = GuardedRE (PEq (ValAt a) (Lit 0))
                (Single (Atom "init" [Num a]))
  , future = \_ → fromRE universe }

increment :: Addr → Int → Pledge GuardedRE ()
increment a maxVal = Pledge
  { ret = ()
  , pre = GuardedRE (PLt (ValAt a) (Lit maxVal)) universe
  , post = fromRE (Single (Atom "inc" [Num a]))
  , future = \_ → fromPPred (PGe (ValAt a) (Lit 0)) }
```

The pre of increment carries $h[a] < M$ (pure Presburger) conjoined with the universal RE. After composing eleven increments against a bound of 10, `normalizePPred` derives the contradiction $h[a] < 10 \wedge \neg(h[a] < 10) \equiv \text{false}$, flagging the overflow before any execution takes place.

Bad programs and residuals.

Bad program	Residual pre (Presburger)	Residual pre (RE)
overflow (11 incs, max=10)	false	$\neg\emptyset$
underflow (dec at zero)	false	$\neg\emptyset$
noInit (skip init)	$\neg\emptyset$	$\emptyset$

6 The WeightedRE Instance

The RE and GRE instances are Boolean: a future condition is either satisfied or violated. The WRE instance generalises membership to a *weight* drawn from a semiring, enabling probabilistic and quantitative obligations to be expressed within the same Composable framework.

6.1 The Semiring Class

```
class (Eq w, Show w) ⇒ Semiring w where
  szero :: w          -- additive identity (blocked path)
  sone  :: w          -- multiplicative identity
  sadd  :: w → w → w
  smul  :: w → w → w
```

Setting $w = \text{Bool}$ (with $\text{sadd} = (\vee)$, $\text{smul} = (\wedge)$) recovers the Boolean RE instance. Two further instances capture quantitative scenarios:

Instance	szero	sadd	smul
Prob (probability)	0	$\min(p_1 + p_2, 1)$	$p_1 \times p_2$
Tropical (min-cost)	$+\infty$	min	+

In the probability semiring, sadd is bounded addition (sum of probabilities capped at 1) and smul is multiplication. In the tropical semiring, sadd is minimum (taking the cheaper alternative) and smul is addition (summing costs along a path).

6.2 The WRE Type and Operators

```
data WRE w
  = WBot                -- weight szero: no accepting path
  | WEps w              -- weight w: accept the empty trace
  | WSingle w Event     -- weight w: accept exactly this event
  | WSeq (WRE w) (WRE w)
  | WAdd (WRE w) (WRE w)
  | WAnd (WRE w) (WRE w)
  | WStar (WRE w)
```

The generalised nullable function accumulates the weight of all accepting paths through the empty trace:

```
wNullable :: Semiring w => WRE w -> w
wNullable (WEps w)      = w
wNullable (WSeq r1 r2)  = smul (wNullable r1) (wNullable r2)
wNullable (WAdd r1 r2)  = sadd (wNullable r1) (wNullable r2)
wNullable (WAnd r1 r2)  = smul (wNullable r1) (wNullable r2)
wNullable (WStar r)     = sone
wNullable _             = szero
```

wDerivative and wSubtraction follow the same structure as their Boolean counterparts, with semiring operations substituted for Boolean ones. The Composable (**WRE** w) instance (enabled by FlexibleInstances) maps concatenation to WSeq, conjunction to WAnd, and subtraction to wSubtraction.

6.3 Derived Forms

```
wTop      :: Semiring w => WRE w
wTop      = WStar (WSingle sone Wildcard) -- sone-weighted anything

wFinally :: Semiring w => w -> Event -> WRE w
wFinally weight e = WSeq wTop (WSingle weight e)

wGlobally :: Semiring w => w -> Event -> WRE w
wGlobally weight e = WStar (WSingle weight e)
```

6.4 Example: Probabilistic Memory (Prob Semiring)

In a probabilistic setting, each `free(a)` has an associated probability of being called. `malloc` carries a future condition asserting that `free(a)` must be called with probability at least p :

```
malloc :: Addr -> Double -> Pledge (WRE Prob) Addr
malloc a p = Pledge
  { ret = a, pre = wTop
```

```
, post  = WSingle sone (Atom "malloc" [Num a])
, future = \r → wFinally (Prob p) (Atom "free" [Num r]) }
```

The residual weight of `wNullable` at the end of a program measures how much of the obligation has been satisfied: a weight of `Prob 1.0` indicates that the obligation is fully discharged; a weight strictly less than 1 quantifies the probability shortfall.

6.5 Example: Min-Cost Scheduling (Tropical Semiring)

Under the tropical semiring, `WSingle (Tropical c) e` assigns cost `c` to emitting event `e`; `wNullable` then returns the minimum total cost of reaching an accepting state:

```
submitJob :: Int → Double → Pledge (WRE Tropical) ()
submitJob jobId cost = Pledge
{ ret = ()
, pre = wTop
, post = WSingle (Tropical cost) (Atom "submit" [Num jobId])
, future = \_ →
    wFinally (Tropical 0) (Atom "complete" [Num jobId]) }
```

Composing several `submitJob` calls yields a future whose `wNullable` is the minimum total cost of completing all submitted jobs. A program that omits a complete event for some job `i` retains `wFinally (Tropical 0) (complete i)` in the residual, and the overall weight is $+\infty$ (i.e. `szero`), flagging the missing completion.

7 The SL Instance

The Composable typeclass is not inherently tied to trace-based specifications. We demonstrate its generality with a fourth instance over *separation-logic heap predicates*, where future conditions describe what heap state must hold upon completion of the computation.

7.1 Structural Parallel with RE

Table 3 shows the role each SL construct plays in the Composable interface.

Table 3. Structural parallel between the four Composable instances.

Operation	RE	GRE	WRE _w	SL
concatenation	$r_1 \cdot r_2$	$(p_1 \wedge p_2, r_1 \cdot r_2)$	<code>WSeq</code>	$P * Q$
conjunction	$r_1 \wedge r_2$	$(p_1 \wedge p_2, r_1 \wedge r_2)$	<code>WAnd</code>	$P \wedge Q$
empty	ε	$(\text{true}, \varepsilon)$	<code>WEps sone</code>	emp
universe	$\neg \emptyset$	$(\text{true}, \neg \emptyset)$	<code>wTop</code>	T
subtraction	Brzozowski quotient	quotient $(\text{RE}) + \wedge (\text{PPred})$	<code>wSubtraction</code>	$P \multimap Q$

The bind formula is unchanged: $\text{pre}(e \gg f) = \text{pre}(e) \wedge (\text{post}(e) \multimap \text{pre}(e_2))$ and $\text{future}(e \gg f) = (\text{post}(e_2) \multimap \text{future}(e)) \wedge \text{future}(e_2)$. Here $\multimap$ is the magic wand (separating implication): $P \multimap Q$ holds of heap h iff for every heap h' disjoint from h satisfying P , the combined heap $h \cup h'$ satisfies Q .

7.2 The SL Predicate Type

```
data SL
= Emp          -- empty heap      (identity for SepStar)
| Top          -- any heap        (identity for Conj)
```

```

| Pure PPred      -- pure Presburger constraint (heap-independent)
| Cell Addr Val   -- singleton: address a holds value v
| SepStar SL SL   -- P * Q separating conjunction
| Conj SL SL      -- P ∧ Q ordinary conjunction
| Wand SL SL      -- P -* Q magic wand (residual)
deriving (Eq, Show)

```

7.3 Separating vs. Ordinary Conjunction

SepStar $P Q$ holds of heap h iff h splits into disjoint parts $h_1 \uplus h_2$ with $P(h_1)$ and $Q(h_2)$; it is the standard separating conjunction of ownership. Conj $P Q$ holds of h iff the *same* heap satisfies both P and Q . Without pure constraints, Conj on purely spatial predicates is either trivial ($\top \wedge P = P$) or unsatisfiable (Cell $l_1 v_1 \wedge$ Cell $l_2 v_2 = \text{false}$ for $l_1 \neq l_2$). The constructor becomes useful when one side is a pure constraint $[\phi] \wedge (\ell \mapsto v)$, combining a heap-independent condition with spatial ownership.

7.4 Presburger Arithmetic for Pure Constraints

Pure constraints are modelled by a small Presburger arithmetic language over heap values. Expressions are linear combinations of heap dereferences and literals; predicates are comparisons and propositional connectives.

```

data PExpr
= Lit Int          -- integer literal
| ValAt Addr       -- h[a]: value at address a
| Add PExpr PExpr  -- e1 + e2
| Mul Int PExpr    -- k * e (scalar only, keeps linearity)

data PPred
= PTrue
| PLt PExpr PExpr  | PLe PExpr PExpr  | PEq PExpr PExpr
| PGt PExpr PExpr  | PGe PExpr PExpr
| PNot PPred       | PAnd PPred PPred

```

Derived connectives (pFalse, pOr) are smart constructors built via De Morgan laws. Pure p holds of any heap for which evalPPred $h p$ is true; it enforces no ownership of cells. The smart constructor pureSL realises the standard SL box notation: pureSL **PTrue** = Emp and pureSL $p =$ Conj (Pure p) Emp for non-trivial p .

7.5 The Composable SL Instance

```

instance Composable SL where
  concatenation = SepStar
  conjunction   = Conj
  empty         = Emp
  universe      = Top

  subtraction Emp q = q          -- emp -* Q = Q
  subtraction Top _ = Emp        -- Top trivially covers any pre
  subtraction p q    = Wand p q  -- general: symbolic wand

```

The two base cases mirror those of the RE instance: $\varepsilon \setminus r = r$ becomes $\mathbf{emp} \multimap Q = Q$ (providing no heap leaves the obligation unchanged), and $\neg\emptyset \setminus r = \varepsilon$ becomes $\top \multimap Q = \mathbf{emp}$ (a $\top$ postcondition covers any precondition, leaving no residual). The general case records the wand symbolically for later evaluation or normalisation.

7.6 Normalisation

normalizeSL applies the standard SL algebraic identities:

Law	Constructor
$\mathbf{emp} * Q = Q, P * \mathbf{emp} = P$	SepStar
$\top \wedge Q = Q, P \wedge \top = P$	Conj
$P \wedge P = P$ (idempotent)	Conj
$\mathbf{emp} \multimap Q = Q$	Wand
$P \multimap \top = \top$	Wand

All rules apply recursively bottom-up.

8 Examples

We demonstrate Pledge across ten use cases spanning all four instances. Each example defines primitive operations annotated with pre, post, and future, composes them in the Pledge monad, and inspects the normalised annotations of the result. A future of $\neg\emptyset$ (or wNullable = some for WRE) and a pre of $\neg\emptyset$ together certify temporal correctness.

8.1 TLS Handshake Lifecycle

The tls library obligation is representative of a broad class of Haskell protocol obligations: a successful handshake must always be paired with a subsequent bye. We model the Context as an integer identifier.

```

type Context = Int

tlsHandshake :: Context → Pledge RE Context
tlsHandshake ctx = Pledge
  { ret    = ctx
  , pre    = universe
  , post   = Single (Atom "handshake" [Num ctx])
  , future = \c → finally (Atom "bye" [Num c]) }

tlsBye :: Context → Pledge RE ()
tlsBye ctx = Pledge
  { ret    = ()
  , pre    = Single (Atom "handshake" [Num ctx])
  , post   = Single (Atom "bye" [Num ctx])
  , future = \_ → universe }

tlsSend :: Context → Pledge RE ()
tlsSend ctx = Pledge
  { ret    = ()
  , pre    = Single (Atom "handshake" [Num ctx])
  , post   = Single (Atom "send" [Num ctx])

```

```

    , future = \_ → universe }

-- Good: handshake followed by data exchange then bye
goodSession = do
  ctx <- tlsHandshake 1
  tlsSend ctx
  tlsBye ctx

-- Bad future: bye omitted; residual = finally(bye(1))
missingBye = do
  ctx <- tlsHandshake 1
  tlsSend ctx

-- Bad future: two contexts opened, only one closed
partialClose = do
  ctx1 <- tlsHandshake 1
  ctx2 <- tlsHandshake 2
  tlsBye ctx1

```

missingBye retains finally(bye(1)) as its residual future, naming the unclosed context precisely. partialClose retains finally(bye(2)): the future condition is data-dependent, so the residual identifies context 2 specifically, not a generic “some context was not closed.”

8.2 MVar Lifecycle

takeMVar returns the taken handle; its data-dependent future condition uses the lambda r so the obligation tracks the *returned* handle. putMVar carries noUntil(putMVar(v), takeMVar(v)) as its future, forbidding a second putMVar(v) until takeMVar(v) resets the guard.

```

type MVar = Int

takeMVar :: MVar → Pledge RE MVar
takeMVar v = Pledge
  { ret = v, pre = universe
  , post = Single (Atom "takeMVar" [Num v])
  , future = \r → finally (Atom "putMVar" [Num r]) }

putMVar :: MVar → Pledge RE ()
putMVar v = Pledge
  { ret = ()
  , pre = Single (Atom "takeMVar" [Num v])
  , post = Single (Atom "putMVar" [Num v])
  , future = \_ → noUntil (Atom "putMVar" [Num v]) (Atom "takeMVar" [Num v]) }

-- Good: data-dependent - obligation discharged on the returned handle
takeAndPut n = do { v <- takeMVar n; putMVar v }

-- Good: re-acquire same MVar after releasing - noUntil is reset by takeMVar
takePutTakePut = do { v <- takeMVar 1; putMVar v; v <- takeMVar 1; putMVar v }

-- Bad future: putMVar(2) obligation remains (MVar 2 never returned)

```



```
missingPut = do { v1 <- takeMVar 1; putMVar v1; _ <- takeMVar 2; return () }

-- Bad pre: takeMVar(1) precondition not met
putWithoutTake = putMVar 1

-- Bad future: double-put immediately collapses future to Bot
doublePutImmediate = do { v <- takeMVar 1; putMVar v; putMVar v }

-- Bad future: double-put with intervening work, still collapses to Bot
doublePutWithWork = do
  { v1 <- takeMVar 1; v2 <- takeMVar 2; putMVar v1; putMVar v2; putMVar v1 }
```

The three examples below return non-unit values, showing that future conditions are tracked regardless of the return type.

```
-- Good: ret = MVar; future = finally(putMVar(1)) - obligation visible in residual
takeReturnHandle :: Pledge RE MVar
takeReturnHandle = takeMVar 1

-- Bad future: ret = (MVar,MVar); future = finally(putMVar(1)) ^ finally(putMVar(2))
takeTwoReturnPair :: Pledge RE (MVar, MVar)
takeTwoReturnPair = do { v1 <- takeMVar 1; v2 <- takeMVar 2; return (v1, v2) }

-- Good: ret = MVar; future = noUntil(putMVar(1),takeMVar(1)) - no pending finally
takePutReturnHandle :: Pledge RE MVar
takePutReturnHandle = do { v <- takeMVar 1; putMVar v; return v }
```

8.3 Mutex / Lock Lifecycle

```
release :: Int → Pledge RE ()
release m = Pledge
  { ret = ()
  , pre  = Single (Atom "acquire" [Num m])
  , post = Single (Atom "release" [Num m])
  , future = universe }

-- Bad future: lock 1 not released
lockLeak = do { acquire 1; acquire 2; release 2 }

-- Bad pre: release without acquire
releaseWithoutAcquire = release 1
```

8.4 Database Transactions

`beginTx` carries a *disjunctive* future condition requiring that the transaction is eventually either committed or rolled back; `commit` and `rollback` each discharge this obligation:

```
beginTx :: Pledge RE ()
beginTx = Pledge
  { ret = (), pre = universe
  , post = Single (Atom "beginTx" []) }
```

```

, future = Or (finally (Atom "commit" []))
              (finally (Atom "rollback" [])) }

commit :: Pledge RE ()
commit = Pledge
  { ret = (), pre = Single (Atom "beginTransaction" [])
  , post  = Single (Atom "commit" [])
  , future = universe }

rollback :: Pledge RE ()
rollback = Pledge
  { ret = (), pre = Single (Atom "beginTransaction" [])
  , post  = Single (Atom "rollback" [])
  , future = universe }

-- Bad future: transaction never closed
openTx = do { beginTx }

-- Good: transaction properly committed
committedTx = do { beginTx; commit }

-- Good: transaction properly rolled back
abortedTx = do { beginTx; rollback }

```

openTx retains the full disjunction $\text{finally}(\text{commit}) \vee \text{finally}(\text{rollback})$ as its residual future. committedTx and abortedTx fully discharge the obligation, yielding $\neg\emptyset$.

RE summary. Table 4 summarises the three RE domains.

Table 4. RE instance: bad programs and residual annotations.

Domain	Bad program	Residual future	Residual pre
Memory	missingFree	finally(free(2))	$\neg\emptyset$
Memory	freeWithoutMalloc	$\neg\emptyset$	$\emptyset$
Memory	doubleFreeImmediate	$\emptyset$	$\neg\emptyset$
Memory	leakAndDoubleFree	$\emptyset$	$\neg\emptyset$
Memory	allocReturnAddr	finally(free(1))	$\neg\emptyset$
Memory	allocTwoReturnPair	finally(free(1)) $\wedge$ finally(free(2))	$\neg\emptyset$
Mutex	acquire 1 leaked	finally(release(1))	$\neg\emptyset$
Mutex	release without acquire	$\neg\emptyset$	$\emptyset$
Transaction	no commit/rollback	finally(commit) $\vee$ finally(rollback)	$\neg\emptyset$

8.5 Bounded Counter (GRE Instance)

The bounded counter (§5) demonstrates that both the Presburger and RE components are necessary. The RE alone would accept overflow sequences; the Presburger predicate alone would accept arbitrarily ordered events.

```
-- Good: init, two increments, one decrement, snapshot
```

```
normalUse :: Pledge GuardedRE ()
normalUse = do
  initCounter 0; increment 0 10; increment 0 10
  decrement 0; snapshot 0

-- Bad: 11 increments against max 10
overflow :: Pledge GuardedRE ()
overflow = do
  initCounter 0
  sequence_ (replicate 11 (increment 0 10))
  snapshot 0
```

For overflow, the combined pre after composing all eleven increments normalises to $[\mathbf{false}] \wedge \neg\emptyset$: the Presburger side collapses to $\mathbf{PFalse}$ while the RE side remains the universal language, pinpointing the arithmetic violation.

GRE *summary*.

Bad program	Residual pre (GRE)	Cause
overflow	$[\mathbf{false}] \neg\emptyset$	$h[0] < 10$ violated
underflow	$[\mathbf{false}] \neg\emptyset$	$h[0] > 0$ violated
noInit	$[\mathbf{true}] \emptyset$	init event missing

8.6 Task Scheduling (WRE Instance)

The task-scheduling example shows that the tropical semiring naturally accumulates minimum costs. Each `submitJob` operation carries both a submission cost and a future condition requiring that every submitted job eventually completes.

```
-- Good: submit two jobs, complete both
goodSchedule :: Pledge (WRE Tropical) ()
goodSchedule = do
  submitJob 1 2.0; submitJob 2 3.0
  completeJob 1; completeJob 2

-- Bad: job 2 never completed
missingComplete :: Pledge (WRE Tropical) ()
missingComplete = do
  submitJob 1 2.0; submitJob 2 3.0
  completeJob 1
```

For `goodSchedule`, `wNullable (evalFuture ...)` returns $\mathbf{Tropical} \ 0$, indicating all obligations are discharged at zero residual cost. For `missingComplete`, the residual future still contains `wFinally ( $\mathbf{Tropical} \ 0$ ) (complete 2)`, and the overall weight is $+\infty$, flagging job 2 as unfinished.

The probability-semiring memory example follows analogously: a `malloc a p` with $p = 0.9$ leaves a residual weight of 0.9 if `free a` is never called, dropping to 0 when it is.

8.7 Heap Memory (SL Instance)

The heap-memory example mirrors the RE memory example but uses heap ownership rather than event traces.

```
alloc :: Addr → Val → Pledge SL ()
```

```

alloc a v = Pledge
  { ret = (), pre = Top, post = Cell a v, future = Top }

free :: Addr → Val → Pledge SL ()
free a v = Pledge
  { ret = (), pre = Cell a v, post = Emp, future = Top }

writeCell :: Addr → Val → Val → Pledge SL ()
writeCell a old new = Pledge
  { ret = (), pre = Cell a old, post = Cell a new, future = Top }

```

Good: `alloc 0 1 >> alloc 1 2`. The combined post is SepStar (Cell 0 1) (Cell 1 2): two disjoint ownerships composed by separating conjunction, correctly reflecting that the two cells reside at distinct addresses.

Bad: `leak (alloc 0 1 >> alloc 1 2 >> free 0 1)`. After freeing only address 0, the post retains Cell 1 2, naming the unreleased ownership.

Bad: `read without ownership (readCell 0 42)`. The pre of the bare read is Cell 0 42, which is never discharged, flagging the missing ownership.

8.8 Bank Account with Presburger Guards (SL Instance)

The bank account example demonstrates Pure Presburger constraints inside Conj.

```

withdraw :: Addr → Val → Val → Pledge SL ()
withdraw a old amount = Pledge
  { ret = ()
  , pre = Conj (Pure (PGe (ValAt a) (Lit amount))) (Cell a old)
  , post = Cell a (old - amount)
  , future = Top }

closeAccount :: Addr → Val → Pledge SL ()
closeAccount a bal = Pledge
  { ret = ()
  , pre = Conj (Pure (PEq (ValAt a) (Lit 0))) (Cell a bal)
  , post = Emp, future = Top }

```

The precondition of `withdraw` is a Conj of a pure Presburger guard ($h[a] \geq \text{amount}$) and a spatial ownership claim (Cell a old). Without the Pure constructor, Conj of two purely spatial predicates would be either trivial or unsatisfiable (see §7); the combination of Pure and Conj is what makes value-dependent preconditions expressible.

Bad: **overdraft.** After `deposit 0 0 100`, the account holds 100. `withdraw 0 100 150` carries $pre = \text{Conj} (\text{Pure} (h[0] \geq 150)) (\text{Cell } 0 \ 100)$. Since $100 < 150$, the pure guard is not satisfiable, and the residual pre of the composed program retains this unsatisfiable constraint.

8.9 Linked List (SL Instance)

A singly-linked-list node occupies two consecutive cells: address a holds the payload value and address $a+1$ holds the next pointer.

```

allocNode :: Addr → Val → Val → Pledge SL ()
allocNode a val next = Pledge
  { ret = (), pre = Top

```

```
, post = SepStar (Cell a val) (Cell (a+1) next)
, future = Top }
```

Two-node list (`allocNode 0 10 2 >> allocNode 2 20 (-1)`). The combined post is a four-cell SepStar:

$$(\text{Cell } 0 \ 10 * \text{Cell } 1 \ 2) * (\text{Cell } 2 \ 20 * \text{Cell } 3 \ -1)$$

This is the canonical representation of a two-node list in separation logic: all four cells are disjointly owned, and the structure of SepStar mirrors the structure of the list.

SL summary. Table 5 summarises the three SL domains. The key difference from the RE instance is that residual annotations are heap predicates rather than traces: a leaked cell appears in post, an unsatisfiable ownership constraint in pre.

Table 5. SL instance: bad programs and residual annotations.

Domain	Bad program	Residual post	Residual pre
Heap memory	Cell 1 2 not freed	Cell 1 2 (leaked)	$\top$
Heap memory	read without ownership	emp	Cell 0 42
Bank account	insufficient balance	Cell 0 -50	unsatisfiable guard
Bank account	close with non-zero balance	emp	unsatisfied $h[0] = 0$
Linked list	read without ownership	emp	Cell 0 99

9 Monad Laws

Because future has type $a \rightarrow \text{eff}$, the monad laws are stated in terms of `evalFuture`, defined as `future e (ret e)`: the future condition evaluated at the computation’s own return value. We write $\widehat{\text{future}}(e)$ for `evalFuture(e)` for brevity. The laws hold for the triple $(\text{ret}, \text{post}, \widehat{\text{future}})$ under the assumption that Composable forms a monoid under $\diamond$ with unit `empty`, and that universe is the unit for $(\wedge \setminus)$. The pre field satisfies the Hoare-rule property (Equation 4) rather than strict monad-law equality; we discuss this in the final paragraph of this section.

Left identity. `pure x >>= f = f x`.

ret: `ret(f x)`. $\checkmark$ *post:* $\varepsilon <> \text{post}(f x) = \text{post}(f x)$. $\checkmark$

For the future, let $fe = f(\text{ret}(\text{pure } x)) = f x$. `future(pure x) =  $\lambda \_ . \neg\emptyset$` , so $\widehat{\text{future}}(\text{pure } x) = \neg\emptyset$.

$$\begin{aligned} \widehat{\text{future}}(\text{pure } x \gg f) &= (\text{post}(fe) \setminus \widehat{\text{future}}(\text{pure } x)) \wedge \widehat{\text{future}}(fe) \\ &= (\text{post}(f x) \setminus \neg\emptyset) \wedge \widehat{\text{future}}(f x) = \varepsilon \wedge \widehat{\text{future}}(f x) = \widehat{\text{future}}(f x). \checkmark \end{aligned}$$

The identity $r \setminus \neg\emptyset = \varepsilon$ follows from the base case $\neg\emptyset \setminus r = \varepsilon$ of `reSubtraction`.

Right identity. `e >>= pure = e`.

ret: `ret(e)`. $\checkmark$ *post:* $\text{post}(e) <> \varepsilon = \text{post}(e)$. $\checkmark$

Let $fe = \text{pure}(\text{ret}(e))$. Then $\text{post}(fe) = \varepsilon$ and

$$\widehat{\text{future}}(fe) = \text{future}(fe)(\text{ret}(fe)) = (\lambda _ . \neg\emptyset)(\text{ret}(e)) = \neg\emptyset.$$

$$\widehat{\text{future}}(e \gg \text{pure}) = (\varepsilon \setminus \widehat{\text{future}}(e)) \wedge \neg\emptyset = \widehat{\text{future}}(e) \wedge \neg\emptyset = \widehat{\text{future}}(e). \checkmark$$

Associativity. For $(e \ggg f) \ggg g = e \ggg (\lambda x. f \ x \ggg g)$, the *ret* and *post* components follow from associativity of $\circ$. For the future, let $r_1 = \text{ret}(e)$, $r_2 = \text{ret}(f \ r_1)$, $r_3 = \text{ret}(g \ r_2)$. Both sides reduce to

$$(\text{post}(g \ r_2) \setminus (\text{post}(f \ r_1) \setminus \widehat{\text{future}}(e))) \wedge \widehat{\text{future}}(f \ r_1) \wedge \widehat{\text{future}}(g \ r_2)$$

by repeated application of Equation (3) and associativity of $\wedge$.

Precondition. The pre field propagates constraints via $\wedge$. When $\text{post}(e)$ fully covers $\text{pre}(e_2)$, Equation (4) gives $\text{pre}(e \ggg f) = \text{pre}(e) \wedge \varepsilon$, which reduces to ε when $\text{pre}(e) = \neg\emptyset$ or $\emptyset$ when $\text{pre}(e)$ is non-nullable, flagging contradictions immediately. The residual $\text{post}(e) \setminus \text{pre}(e_2)$ records precisely which precondition obligations remain unmet.

10 Related Work

Pre/post-condition contracts in Haskell. *Liquid Haskell* [Vazou et al. 2014] expresses pre/post-conditions as refinement types verified by an SMT solver. Findler and Felleisen [Findler and Felleisen 2002] establish a runtime contract discipline for higher-order functions. QuickCheck [Claessen and Hughes 2000] supports conditional properties via $\Rightarrow$. Swierstra and Baanen [Swierstra and Baanen 2019] embed Hoare triples as an indexed monad, propagating pre/post annotations through monadic bind analogously to Pledge. Pledge complements these by adding a *temporal* dimension through the future condition, without requiring type-system extensions.

Future conditions and temporal verification. ProveNFix [Song et al. 2024] encodes temporal properties as regular expressions to guide automated program repair. Pledge is complementary in orientation: rather than repairing programs after the fact, it carries temporal obligations as first-class annotations so that violations are identified statically, prior to any execution. Song et al. [2025] propose future conditions as a first-class extension to pre/post-condition contracts, formalising the notion that an operation may discharge temporal obligations onto its continuation. Pledge realises this idea as a Haskell library, contributing a monadic interface, derivative-based propagation, and four concrete Composable instances that the formal system leaves unaddressed.

Parameterised and graded monads. Atkey [Atkey 2009] introduces parameterised monads $T(S_1, S_2, A)$ for Hoare-logic-style pre/post state tracking. Katsumata [Katsumata 2014] generalises this to graded monads over a monoid; Orchard and Wu [Orchard and Yoshida 2016] show effect systems and session types are instances. Pledge is related but orthogonal: rather than indexing by effect *type* at the type level, we carry an explicit obligation as *future* $:: a \rightarrow \text{eff}$ at the value level, preserving compatibility with the standard Monad class, **do**-notation, and the `mtl` stack.

Resource management in Haskell. `bracket` lexically scopes cleanup but cannot propagate obligations beyond the lexical scope. `ResourceT` [Snoyman 2011] threads a finaliser queue through every bind, discharging cleanup when the block exits. `Acquire` and `Managed` generalise this to an applicative/monadic interface. All three handle the *cleanup* half of a resource obligation but cannot express *protocol* or *quantitative* obligations, which require explicit future-condition propagation across bind steps.

Linear types. Linear Haskell [Bernardy et al. 2018] introduces multiplicity annotations enforced by GHC since version 9.0; `linear-base` [Tweag and contributors 2021] provides a `RIO` monad with linear file handles, statically guaranteeing every opened handle is consumed exactly once. Linear types enforce *structural* obligations (use exactly once), whereas future conditions enforce *temporal* obligations (use in a specified order, or with a specified probability). A hybrid combining linear types for uniqueness with Pledge for temporal ordering could achieve stronger guarantees than either alone.

Effect systems and algebraic effect handlers. Algebraic effect handlers [Pretnar 2015] decompose effectful computations into an operation set and a separate handler; resource lifecycle obligations are typically encoded in the handler’s state machine. Polysemy’s Scoped effect attaches resource creation and cleanup to a lexical scope threaded through the effect row. Pledge takes the complementary route: obligations are values in the annotation fields, composable across any handler.

Session types. Session types [Gay and Hole 2005; Honda 1993] enforce communication protocols at the type level. The sessiontypes library [van Walree 2017] realises this via an indexed monad $\text{STTerm } m \rightarrow s \rightarrow s' \rightarrow a$; the type checker rejects programs that violate the protocol order. Pledge is more lightweight: it operates within the standard Monad class, admits quantitative and heap-ownership obligations that session types do not, and is applicable beyond message-passing.

Typestate. Typestate systems [Bierhoff and Aldrich 2007; Strom and Yemini 1986] encode resource lifecycle protocols in the type system. Brady’s ST monad encoding in Idris [Brady 2013] demonstrates full compile-time enforcement in a dependently typed language. Pledge achieves comparable expressiveness at the library level using standard Haskell type classes, at the cost of moving checking to annotation-inspection rather than type-inference time.

Runtime monitoring. Runtime verification [Leucker and Schallhart 2009] checks temporal properties after execution. Pledge takes a *static* approach: future conditions are propagated before any execution, and the LTL_f embedding eliminates automaton construction.

Brzozowski derivatives. Derivatives of regular expressions [Brzozowski 1964] have found application in parsing [Might et al. 2011; Owens et al. 2009] and model checking. Equation (1) extends this to complement via $\partial_a(\neg r) = \neg(\partial_a r)$; our subtraction operator is a novel application to future-condition propagation in a monadic setting.

11 Discussion and Future Work

Decidability. Equality of normalised regular expressions is decidable, so the fixed-point computation in subtraction terminates whenever the derivative sequence is finite, which holds for the regular languages used in our examples. Extending to context-free or ω -regular specifications remains future work.

Precondition monad laws. The pre field satisfies the Hoare-rule property rather than strict monad-law equality (Section 9). A natural refinement is to make pre a separate, non-monadic annotation computed by a static analysis pass over the Pledge term, separating the behavioural monad from the precondition contract checker.

Integration with effect handlers. Lifting Pledge into an effect-handler framework (effectful or polysemy) would allow temporal specifications to compose alongside algebraic effect rows. A lightweight *shadow* approach requires no deep integration: keep the Pledge RE computation as a pure specification evaluated statically, while a separate Eff es computation carries out the real effects, with assertions linking the two at the top level:

```
main = do
  assert (normalize (pre      spec) == top) "precondition_violated"
  assert (normalize (evalFuture spec) == top) "future_obligation_unmet"
  runEff impl
```

Static checking is entirely pure; no changes to the effect stack are required.

Closed-world alphabet assumption. `firstWith` uses $\Sigma_0 = \text{atoms}(r_1) \cup \text{atoms}(r_2)$ as its working alphabet. This is sound under the *closed-world assumption* that every relevant event is mentioned in the two operands. A fully open-world implementation would accept an explicit Σ declaration, enabling complement over an unrestricted domain.

Connection to indexed and graded monads. Because `future :: a → eff` indexes the obligation by the return value, `Pledge` echoes the indexed/graded-monad literature where types are indexed by computational effects; here the index is a return value instead of an effect row.

Solver-backed discharge of residual conditions. Residual annotations are currently surfaced as symbolic expressions. A natural extension would connect them to an automatic solver: RE residuals checked by NFA/DFA construction or a BDD library; SL predicates with Presburger guards discharged by Z3 or CVC5, reporting concrete counterexample traces or unsatisfiable heaps.

Parallel composition. `Pledge` is inherently sequential; concurrent programs require a parallel combinator $e_1 \parallel e_2$. In the RE instance this corresponds to the shuffle product $r_1 \sqcup r_2$; the SL instance admits a parallel connective via the concurrent separation logic frame rule [O’Hearn 2007]. Extending `Composable` with a `parallel` operation is a natural next step.

Quantitative future conditions. A natural further direction is to connect WRE to probabilistic temporal logics (PCTL) or quantitative Kleene algebras; in the SL instance, a quantitative analogue could express amortised resource bounds or expected heap usage.

12 Conclusion

We have presented *Pledge*, a Haskell library that augments effectful computations with *future conditions*: annotations that constrain what the remainder of a program must accomplish. Four design decisions underpin the approach.

First, the `Composable` typeclass abstracts the specification algebra into five operations: concatenation, conjunction, subtraction, and their respective identity elements. It admits four independent instances that all share the same monadic bind formula. This uniformity demonstrates that trace-based, arithmetic, quantitative, and heap-ownership obligations are not merely analogous, but structurally identical at the algebraic level.

Second, the RE instance incorporates complement as a first-class constructor, exploiting the single law $\partial_a(\neg r) = \neg(\partial_a r)$ to support globally assertions and a direct embedding of LTL_f , without any automaton construction.

Third, the GRE instance pairs each RE with a Presburger arithmetic predicate, so that a single annotation simultaneously enforces a protocol ordering and a numeric bound; the `normalizePPred` procedure eagerly introduces the canonical `PFalse` constructor, identifying syntactically unsatisfiable constraints before any solver call is issued.

Fourth, the WRE instance generalises Boolean membership to a semiring weight, unifying probabilistic future conditions under the probability semiring and min-cost scheduling obligations under the tropical semiring within a single framework. The SL instance further demonstrates the breadth of the `Composable` interface by instantiating separating conjunction as sequential composition and magic wand as subtraction, with `Pure PPred` inside `Conj` making value-dependent preconditions such as “balance $\geq$ amount” expressible as first-class heap predicates.

The resulting system is lightweight (no language extensions beyond standard Haskell type classes), composable via a standard `Monad` instance, and diagnostically precise across all four dimensions: a residual future identifies unmet trace obligations; a `PFalse` in pre flags an arithmetic violation; a semiring weight of `szero` signals an unfulfilled quantitative obligation; and a leaked `Cell` in a residual post exposes a heap-ownership error.

References

- Robert Atkey. 2009. Parameterised Notions of Computation. *Journal of Functional Programming* 19, 3–4 (2009), 335–376. doi:10.1017/S095679680900728X
- Jean-Philippe Bernardy, Mathieu Boespflug, Ryan R. Newton, Simon Peyton Jones, and Arnaud Spiwack. 2018. Linear Haskell: Practical Linearity in a Higher-Order Polymorphic Language. *Proceedings of the ACM on Programming Languages* 2, POPL, Article 5 (2018), 29 pages. doi:10.1145/3158093
- Kevin Bierhoff and Jonathan Aldrich. 2007. Modular Typestate Checking of Aliased Objects. In *Proceedings of the 22nd Annual ACM SIGPLAN Conference on Object-Oriented Programming Systems and Applications (OOPSLA)*. 301–320. doi:10.1145/1297027.1297050
- Edwin Brady. 2013. Idris, a General-Purpose Dependently Typed Programming Language: Design and Implementation. *Journal of Functional Programming* 23, 5 (2013), 552–593. doi:10.1017/S095679681300018X
- Janusz A. Brzozowski. 1964. Derivatives of Regular Expressions. *J. ACM* 11, 4 (1964), 481–494. doi:10.1145/321239.321249
- Koen Claessen and John Hughes. 2000. QuickCheck: A Lightweight Tool for Random Testing of Haskell Programs. In *Proceedings of the 5th ACM SIGPLAN International Conference on Functional Programming (ICFP)*. 268–279. doi:10.1145/351240.351266
- Robert Bruce Findler and Matthias Felleisen. 2002. Contracts for Higher-Order Functions. In *Proceedings of the 7th ACM SIGPLAN International Conference on Functional Programming (ICFP)*. 48–59. doi:10.1145/581478.581484
- Simon J. Gay and Malcolm Hole. 2005. Subtyping for Session Types in the Pi Calculus. *Acta Informatica* 42, 2-3 (2005), 191–225. doi:10.1007/s00236-005-0177-z
- Kohei Honda. 1993. Types for Dyadic Interaction. In *Proceedings of the 4th International Conference on Concurrency Theory (CONCUR) (Lecture Notes in Computer Science, Vol. 715)*. 509–523. doi:10.1007/3-540-57208-2_35
- Shin-ya Katsumata. 2014. Parametric Effect Monads and Semantics of Effect Systems. *ACM SIGPLAN Notices* 49, 1 (2014), 633–646. doi:10.1145/2578855.2535846
- Martin Leucker and Christian Schallhart. 2009. A Brief Account of Runtime Verification. *Journal of Logic and Algebraic Programming* 78, 5 (2009), 293–303. doi:10.1016/j.jlap.2008.08.004
- Matthew Might, David Darais, and Daniel Spiewak. 2011. Parsing with Derivatives: A Functional Pearl. In *Proceedings of the 16th ACM SIGPLAN International Conference on Functional Programming (ICFP)*. 189–195. doi:10.1145/2034773.2034801
- Peter W. O’Hearn. 2007. Resources, Concurrency, and Local Reasoning. *Theoretical Computer Science* 375, 1–3 (2007), 271–307. doi:10.1016/j.tcs.2006.12.035
- Dominic Orchard and Nobuko Yoshida. 2016. Effects as Sessions, Sessions as Effects. In *Proceedings of the 43rd ACM SIGPLAN-SIGACT Symposium on Principles of Programming Languages (POPL)*. 568–581. doi:10.1145/2837614.2837634
- Scott Owens, John Reppy, and Aaron Turon. 2009. Regular-expression Derivatives Re-examined. *Journal of Functional Programming* 19, 2 (2009), 173–190. doi:10.1017/S0956796808007090
- Matija Pretnar. 2015. An Introduction to Algebraic Effects and Handlers. Invited tutorial paper. *Electronic Notes in Theoretical Computer Science* 319 (2015), 19–35. doi:10.1016/j.entcs.2015.12.003
- Michael Snoyman. 2011. conduit: Streaming Data Processing. <https://hackage.haskell.org/package/conduit>. Haskell package on Hackage; introduces the ResourceT transformer.
- Yahui Song, Darius Foo, and Wei-Ngan Chin. 2025. Specifying and Verifying Future Conditions. In *Static Analysis — 32nd International Symposium, SAS 2025 (Lecture Notes in Computer Science, Vol. 16100)*. Springer, 90–112. doi:10.1007/978-3-032-07106-4_5
- Yahui Song, Xiang Gao, Wenhua Li, Wei-Ngan Chin, and Abhik Roychoudhury. 2024. ProveNFix: Temporal Property-Guided Program Repair. *Proceedings of the ACM on Software Engineering* 1, FSE, Article 11 (2024), 23 pages. doi:10.1145/3643737
- Robert E. Strom and Shaula Yemini. 1986. Typestate: A Programming Language Concept for Enhancing Software Reliability. *IEEE Transactions on Software Engineering* 12, 1 (1986), 157–171. doi:10.1109/TSE.1986.6312929
- Wouter Swierstra and Tim Baanen. 2019. A Predicate Transformer Semantics for Effects (Functional Pearl). *Proceedings of the ACM on Programming Languages* 3, ICFP, Article 103 (2019), 26 pages. doi:10.1145/3341707
- Tweag and contributors. 2021. linear-base: Standard Library for Linear Types in Haskell. <https://github.com/tweag/linear-base>.
- Ferdinand van Walree. 2017. sessiontypes: Session Types for Haskell. <https://hackage.haskell.org/package/sessiontypes>.
- Niki Vazou, Eric L. Seidel, Ranjit Jhala, Dimitrios Vytiniotis, and Simon Peyton Jones. 2014. Refinement Types for Haskell. In *Proceedings of the 19th ACM SIGPLAN International Conference on Functional Programming (ICFP)*. 269–282. doi:10.1145/2628136.2628161